6 You are given that $\mathbf{v} = 2\mathbf{a} + 3\mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are the position vectors

$\mathbf{a} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 6 \end{pmatrix}$.

(a) Determine the magnitude of $\mathbf{v}$. **[3]**

(b) Determine the angle between $\mathbf{v}$ and the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. **[2]**